

Introduction to Microelectronics: From Silicon to Computer Components



Dr. Nikolaos Athanasios Anagnostopoulos

- Every Thursday from 16.00 to 18.00 in (ITZ) SR 004, from 16.04.2026 until and including 16.07.2026.
- Attendance is advised.
- If the attendees cannot fit in the current classroom, we will try to find a solution.

The following topics will be covered:

- Basic concepts of microelectronics (examples of electronic systems, electronics vs. microelectronics, etc.)
- Short introduction to the physics of microelectronics (charge carriers, the pn junction, the need for conductors and semiconductors, silicon and other semiconducting materials)
- Diodes and their applications
- MOS transistors (PMOS, NMOS, CMOS), their principles of operation, and their applications

The following topics will also be covered:

- CMOS amplifiers
- Operational amplifiers, modes of operation (noninverting amplifier, inverting amplifier, integrator and differentiator, voltage adder), and their applications
- Design principles of simple computer components based on microelectronics (memory cells, registers, logical gates)
- Basic digital circuits and practical applications

The following more advanced topics will also be covered:

- Circuit design, correction, and theoretical testing using artificial intelligence
- Microelectronics beyond silicon: connection to nanoelectronics and post-CMOS technologies

- **Electronics**

- Scientific field
- Applies principles of physics
- Concerns devices that manipulate electrons and other electrically charged particles

- **Microelectronics**

- Subfield of electronics
- Very small electronic components
- Micrometre-scale or smaller
- Semiconductor materials



<https://cdn.britannica.com/50/114750-050-06EEB5F0/compound-microscope.jpg>

- **Nanoelectronics**

- Subfield of electronics
- Nanotechnology in electronics
- Inter-atomic interactions
- Quantum mechanical properties



https://upload.wikimedia.org/wikipedia/commons/c/c5/Electron_Microscope.jpg

- **Electricity**

- Physical phenomenon
- Presence and motion of electric charge

- **Analogy to water**

- Static charge: Glass of water
- Moving charge (thunder): Waterfall



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- **Analogy to water**
 - Electronics: Water dispensers
 - Microelectronics: Water syringe
 - Nanoelectronics: Dentist's water syringe

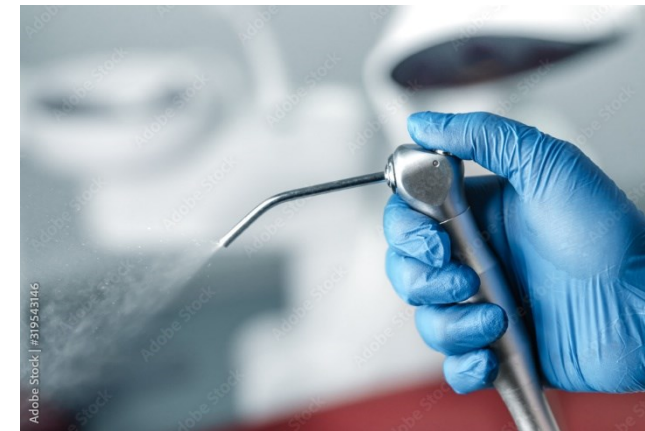


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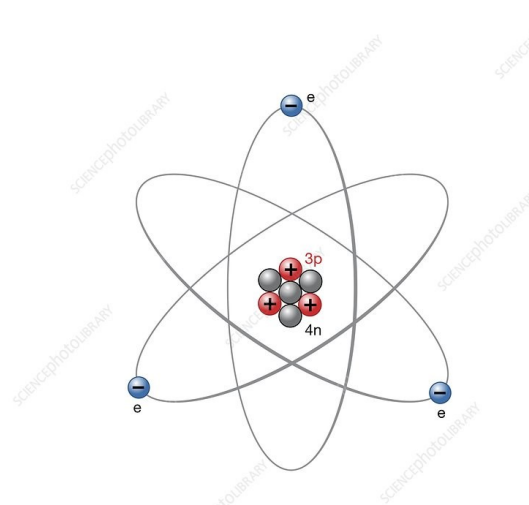
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- Since it has been mentioned that microelectronics concern mostly semiconductor materials and that electronics are based on electrons and other electrically charged particles, we will now need to examine the basics of semiconductors and their electron configurations...



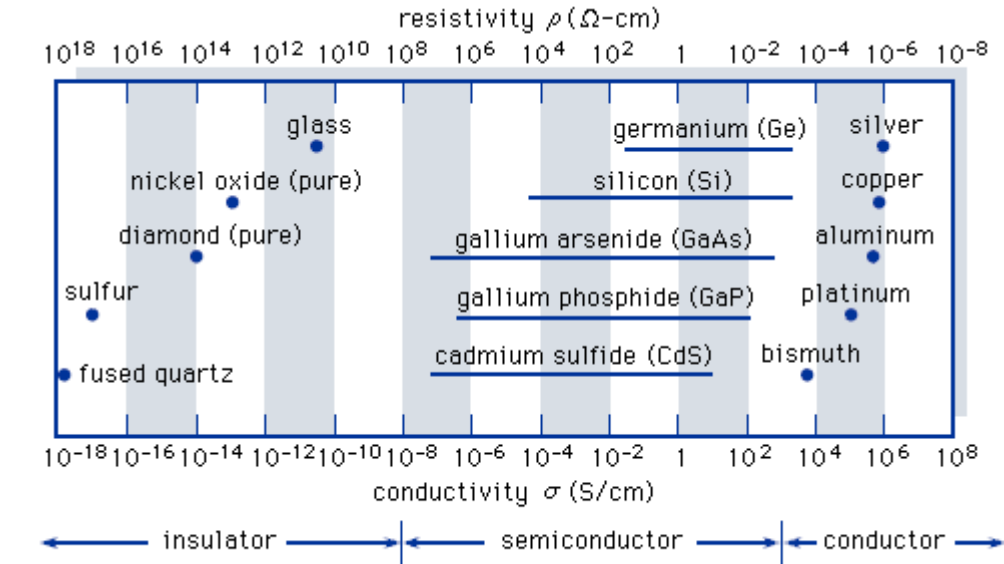
Lithium is a metal and thus a conductor and not a semiconductor. Its ability to form reversible layers between the layers of other layered materials, which is called intercalation, is used in batteries.

https://media.sciencephoto.com/image/c0507593/800wm/C0507593-Lithium_atom_illustration.jpg

- Materials, including pure elements, can be
 - Conductors (low resistivity)
 - Semiconductors (medium resistivity)
 - Insulators (high resistivity)

	IIIA	IVA	VA	VIA
	5 B Boro	6 C Carbonio	7 N Azoto	8 O Ossigeno
	13 Al Alluminio	14 Si Silicio	15 P Fosforo	16 S Zolfo
IIIB	30 Zn Zinco	31 Ga Gallio	32 Ge Germanio	33 As Arsenico
	48 Cd Cadmio	49 In Indio	50 Sn Stagno	51 Sb Antimonio
	80 Hg Mercurio	81 Tl Titanio	82 Pb Piombo	83 Bi Bismuto
			84 Po Polonio	

Conduttori	$\rho < 10^{-5} \Omega m$	(rame: $3 \cdot 10^{-8} \Omega m$)
Semiconduttori	$10^{-5} < \rho < 10^3 \Omega m$	(silicio: $2300 \Omega m$)
Isolanti	$\rho > 10^3 \Omega m$	(diamante: $10^{14} \Omega m$)

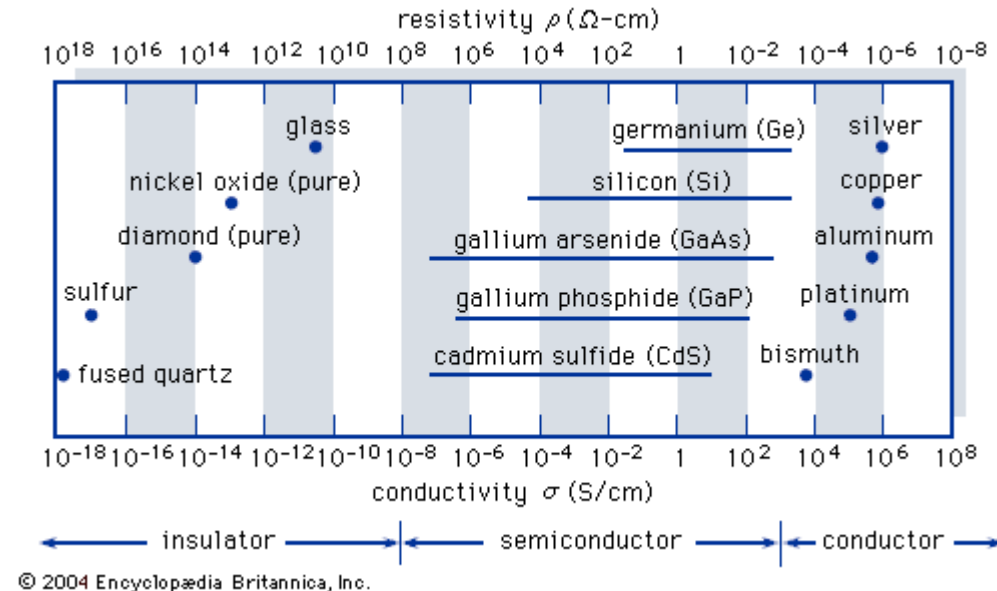


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<https://www.britannica.com/science/semiconductor>

<https://it.wikibooks.org/wiki/File:Semiconductors.png>

- Semiconductors act as bad conductors or as very bad conductors
- Thus, semiconductors can be utilized to control the flow of electric charge
- Silicon is used abundantly as a semiconductor and most electronics are utilizing it
- Silicon (Si)
 - is abundant in the Earth's crust (sand is most often made of quartz (SiO_2))
 - can form high-quality insulating oxide layers (SiO_2) – glass is very often silicon dioxide
 - is compatible with various fabrication processes
 - has relatively low resistivity (in comparison to other semiconductors)



<https://www.britannica.com/science/semiconductor>

The Periodic Table of Elements



PERIODIC TABLE OF ELEMENTS

GROUP	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
PERIOD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H Hydrogen 1.008																	He Helium 4.003
2	Li Lithium 6.94	Be Beryllium 9.012											B Boron 10.81	C Carbon 12.01	N Nitrogen 14.01	O Oxygen 16.00	F Fluorine 19.00	Ne Neon 20.18
3	Na Sodium 22.99	Mg Magnesium 24.31											Al Aluminum 26.98	Si Silicon 28.09	P Phosphorus 30.97	S Sulfur 32.06	Cl Chlorine 35.45	Ar Argon 39.95
4	K Potassium 39.10	Ca Calcium 40.08	Sc Scandium 44.96	Ti Titanium 47.88	V Vanadium 50.94	Cr Chromium 52.00	Mn Manganese 54.94	Fe Iron 55.85	Co Cobalt 58.93	Ni Nickel 58.69	Cu Copper 63.55	Zn Zinc 65.39	Ga Gallium 69.72	Ge Germanium 72.64	As Arsenic 74.92	Se Selenium 78.96	Br Bromine 79.90	Kr Krypton 83.79
5	Rb Rubidium 85.47	Sr Strontium 87.62	Y Yttrium 88.91	Zr Zirconium 91.22	Nb Niobium 92.91	Mo Molybdenum 95.94	Tc Technetium (98)	Ru Ruthenium 101.1	Rh Rhodium 101.1	Pd Palladium 106.4	Ag Silver 107.9	Cd Cadmium 112.4	In Indium 114.8	Sn Tin 118.7	Sb Antimony 121.8	Te Tellurium 127.6	I Iodine 126.9	Xe Xenon 131.3
6	Cs Cesium 132.9	Ba Barium 137.3	57-71 Lanthanides	Hf Hafnium 178.5	Ta Tantalum 180.9	W Tungsten 183.8	Re Rhenium 186.2	Os Osmium 190.2	Ir Iridium 192.2	Pt Platinum 195.1	Au Gold 197.0	Hg Mercury 200.6	Tl Thallium 204.4	Pb Lead 207.2	Bi Bismuth 209.0	Po Polonium (209)	At Astatine (210)	Rn Radon (222)
7	Fr Francium (223)	Ra Radium (226)	89-103 Actinides	Rf Rutherfordium (261)	Db Dubnium (261)	Sg Seaborgium (266)	Bh Bohrium (264)	Hs Hassium (277)	Mt Meitnerium (268)	Ds Darmstadtium (281)	Rg Roentgenium (281)	Cn Copernicium (285)	Nh Nihonium (286)	Fl Flerovium (289)	Mc Moscovium (288)	Lv Livermorium (293)	Ts Tennessine (294)	Og Oganesson (294)
	La Lanthanum 138.9	Ce Cerium 140.1	Pr Praseodymium 140.9	Nd Neodymium 144.2	Pm Promethium (145)	Sm Samarium 150.4	Eu Europium 152.0	Gd Gadolinium 157.3	Tb Terbium 158.9	Dy Dysprosium 162.5	Ho Holmium 164.9	Er Erbium 167.3	Tm Thulium 168.9	Yb Ytterbium 173.0	Lu Lutetium 175.0			
	Ac Actinium (227)	Th Thorium 232.0	Pa Protactinium 231.0	U Uranium 238.0	Np Neptunium (237)	Pu Plutonium (244)	Am Americium (243)	Cm Curium (247)	Bk Berkelium (247)	Cf Californium (251)	Es Einsteinium (252)	Fm Fermium (257)	Md Mendelevium (258)	No Nobelium (259)	Lr Lawrencium (262)			

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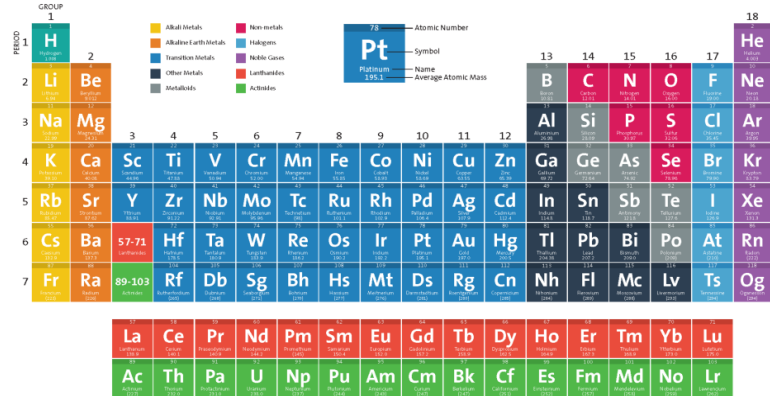
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The Periodic Table of Elements

- The elements on the periodic table are roughly placed based on the number of their protons, which for atoms that are not charged is equal to the number of their electrons.
- Charged atoms contain extra electrons or miss some electrons, and are called ions.
- In fact, the placement of the elements on the periodic table is exactly dependent on the electrons that their non-ionized atoms contain.
- Electrons are ordered into orbitals, which are sequentially filled, and which determine the characteristics of the relevant atoms and, thus, the characteristics of elements.

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PERIODIC TABLE OF ELEMENTS



Legend:

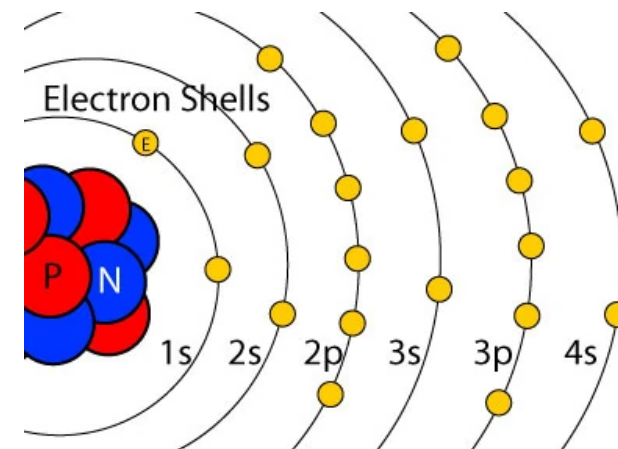
- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Other Metals
- Non-metals
- Halogens
- Noble Gases
- Actinides

Legend:

- Atomic Number
- Symbol
- Name
- Average Atomic Mass

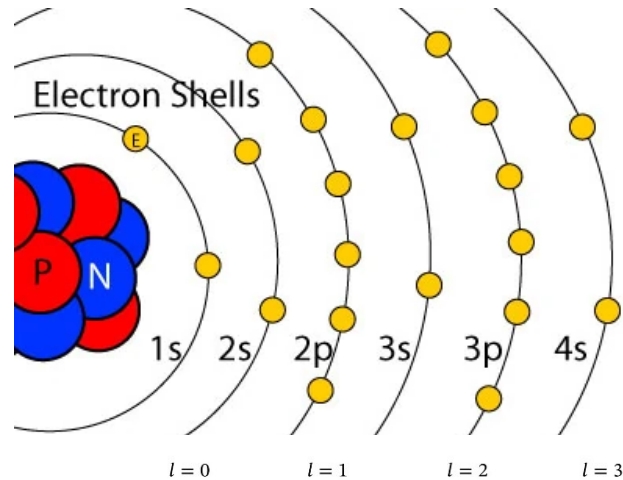
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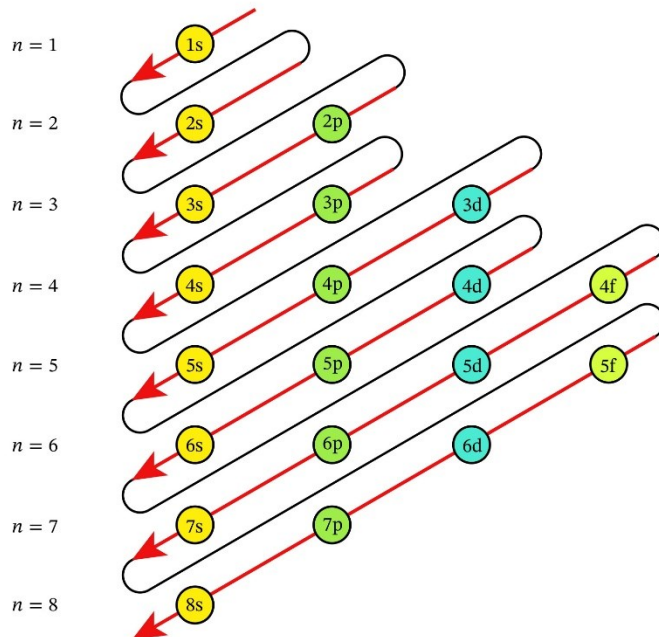


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The Periodic Table of Elements: Orbitals

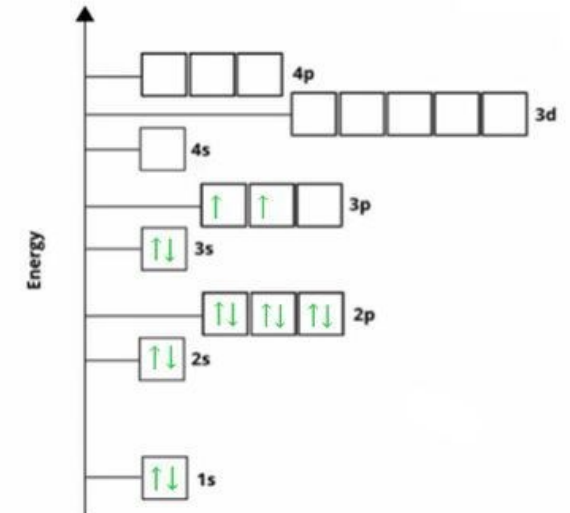
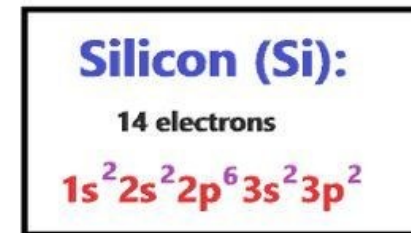


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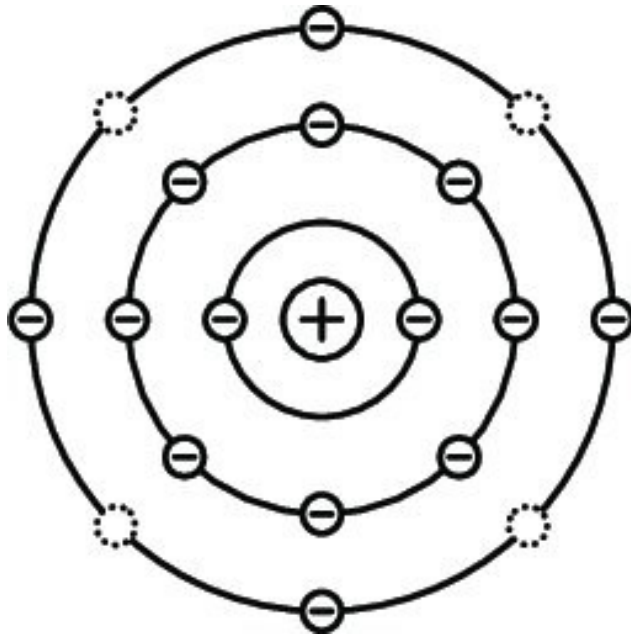
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Electron configuration vs Orbital diagram for Silicon

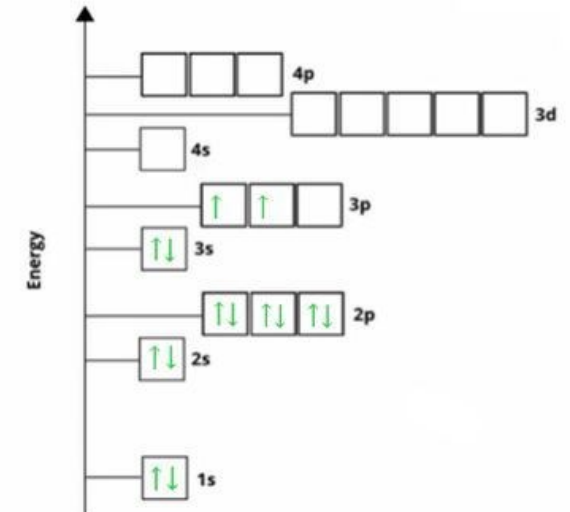
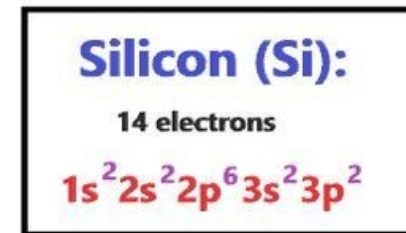


<https://www.pinterest.ca/pin/silicon-si-orbital-diagram-electron-configuration-and-valence-electrons--644437027940039345/>

- Silicon has 4 electrons in its outermost (3^{rd}) electron shell, which would be filled with 8 electrons. At the same time, this means that it has 2 electrons in its outermost (3p) orbital, which would be filled with 6 electrons.



Electron configuration vs Orbital diagram for Silicon



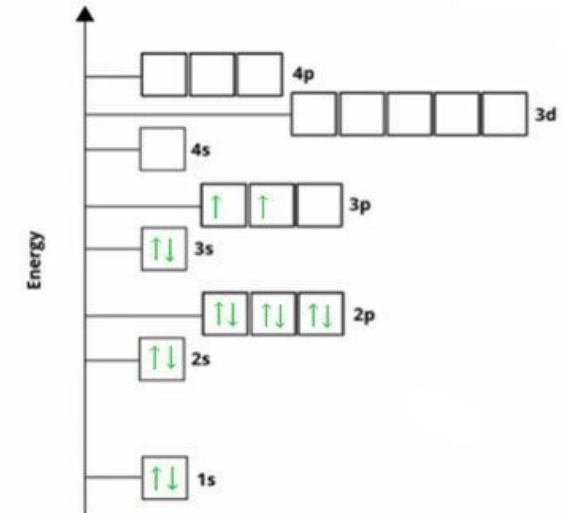
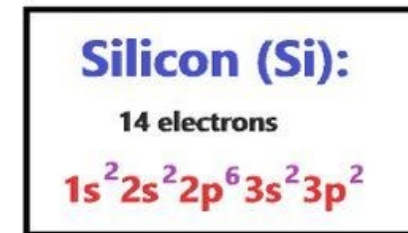
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<https://www.pinterest.ca/pin/silicon-si-orbital-diagram-electron-configuration-and-valence-electrons--644437027940039345/>

- Silicon has 4 electrons in its outermost (3rd) electron shell and 4 electron holes. At the same time, this means that it has 2 electrons in its outermost (3p) orbital and 4 electron holes. This means that it has a tendency both towards attracting and towards expelling electrons to reach a more stable electron configuration.

Valence and Conduction Bands: Silicon

Electron configuration vs Orbital diagram for Silicon

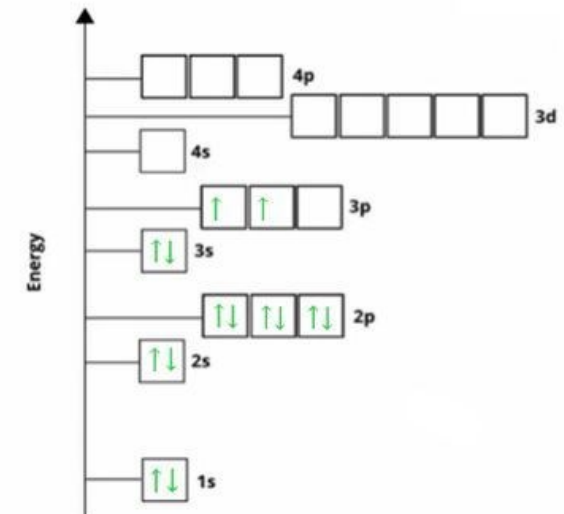
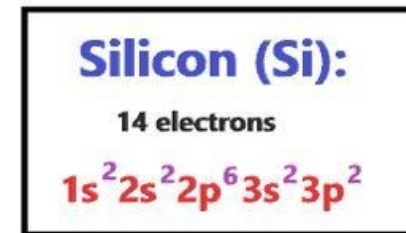


<https://www.pinterest.ca/pin/silicon-si-orbital-diagram-electron-configuration-and-valence-electrons--644437027940039345/>

- Orbitals indicate energy bands for the electrons moving around the nucleus, the core, that is comprised of protons and neutrons.
- To this end, the outermost electron orbitals that electrons actually occupy in a stable non-excited atom form the valence band. The valence band is the band of electron orbitals that electrons can jump out of, moving into the conduction band, when excited.
- The outer orbitals that are not occupied by electrons when the relevant atom is not excited form the conduction band. Moreover, an electron jump to an orbital that is too far from the nucleus will lead to its complete escape and thus to ionization.
- The energy gap between the valence and the conduction bands constitutes the bandgap, and is essentially the energy gap between the last orbitals that are occupied by electrons in the non-excited state and the next non-occupied orbitals.
- Thus, in the case of silicon, its four electrons at the 3s and 3p orbitals constitute electrons in the valence band that can move to the conduction band at the 4s, 3d, or even 4p orbitals.

Electron configuration vs Orbital diagram for Silicon

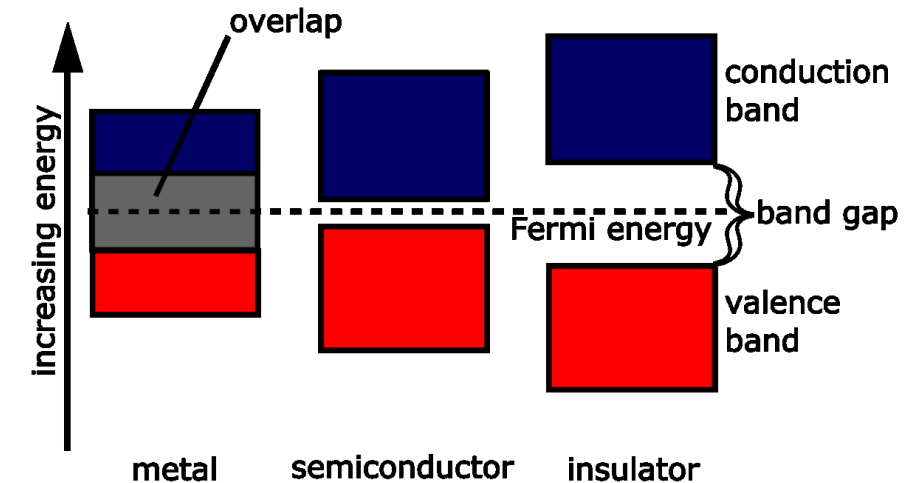
- The valence band consists of the energy level(s) (of the orbitals) of the outermost electrons of an atom at its non-excited state, which (the electrons) can be freed from the atom to turn it into an ion.
- In a broad sense, the valence band consists of the outermost orbitals/shells that contain electrons that can be excited (to the conduction band) and freed from the atom, to turn it into an ion.
- The conduction band consists of the energy level(s) that excited (and/or potentially freed) electrons can exhibit.
- In a broad sense, it contains the (real or hypothetical) orbitals of excited (and/or freed) electrons at the moment of their excitation and/or release from the atom whose influence they are escaping.



<https://www.pinterest.ca/pin/silicon-si-orbital-diagram-electron-configuration-and-valence-electrons--644437027940039345/>

Valence and Conduction Bands

- The relation between the valence and the conduction bands defines whether a material will be a conductor, a semiconductor, or an insulator.
 - In a conductor (which is most often a metal), the valence and the conduction bands' energy levels overlap.
 - In a semiconductor, the two bands have a small energy level gap (a band gap).
 - In an insulator, the bandgap is such that only rarely allows an electron to be excited such that it moves to the conduction band, and thus the relevant material does not ...conduct.
-
- It needs to be noted that all this is a simplification, as the actual orbitals are often combined (hybridized) with each other, leading to slightly different energy levels



https://energyeducation.ca/encyclopedia/Band_gap